Logistic Regression

David Puelz
The University of Austin

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The OkCupid data set

- The OkCupid data set contains information about 59946 profiles from users of the OkCupid online dating service.
- We have data on user age, height, sex, income, sexual orientation, education level, body type, ethnicity, and more.
- OkCupid often publishes their own analyses—see `https://theblog.okcupid.com/tagged/data`.
- Let's see if we can predict the sex/gender of the user based on their height.

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The Y variable here is **categorical** (two levels—everyone in this data set is either labeled male or female), so regular linear regression might not be the best choice here.

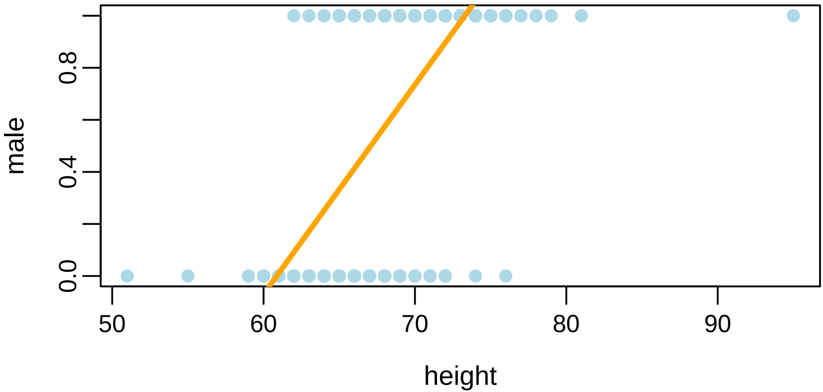
But what if we just do it anyway?

Let's first create a dummy variable to convert sex to a quantitative dummy variable:

```
profiles <- profiles %>% mutate(male = ifelse(sex == "m", 1,
```

We could do this with 1 representing either male or female (it wouldn't matter).

But what if we just do it anyway?

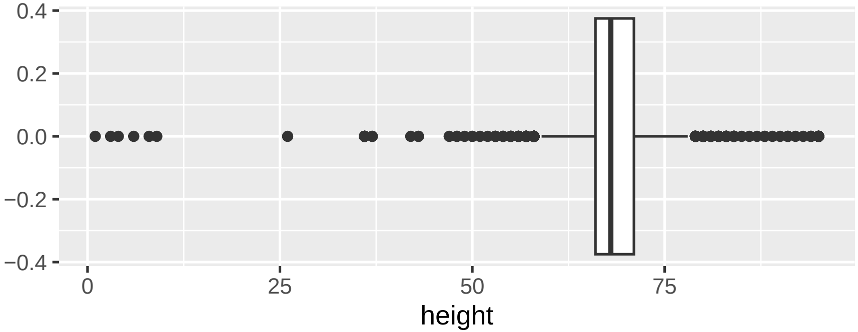


A line is a spectacularly bad fit to this data. And what does it mean to predict that male $= 0.7$ (or 1.2)?

Cleaning the data

There are definitely some weird values for height:

```r
ggplot(profiles, aes(x = height)) + geom_boxplot()
```



Cleaning the data

Let's consider only heights between 55 and 80 inches (4'7" and 6'8"), inclusive. This excludes only 117 cases out of 59946.

```
my.profiles <- profiles %>% filter(height >= 55 & height <= 8
```

The idea behind logistic regression

- Instead of predicting whether someone is male, let's predict the *probability* that they are male
- In logistic regression, one level of Y is always called "success" and the other "failure." Since $Y = 1$ for males, we designate males as "success."
- Let's fit a curve that is always between 0 and 1.

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- In general: The odds of something that happens with probability p are $p/(1-p)$

The logistic regression model

Logistic regression models the **log odds** of success p as a linear function of X:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X$$

This fits an S-shaped curve to the data.

Let's try it

```
model <- glm(male ~ height, data=my.profiles, family=binomial
summary(model)
```

Let's try it

```
Call:
glm(formula = male ~ height, family = binomial, data = my.profiles)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -44.448609   0.357510  -124.3   <2e-16 ***
height        0.661904   0.005293   125.1   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 80654  on 59825  degrees of freedom
Residual deviance: 44637  on 59824  degrees of freedom
AIC: 44641

Number of Fisher Scoring iterations: 6
```

How to interpret the curve?

The regression output tells us that our prediction is

$$\text{log(odds)} = \log\left(\frac{P(\text{male})}{1 - P(\text{male})}\right) = -44.45 + 0.66 \cdot \text{height}.$$

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Solving for $P(\text{male})$:

$$\widehat{P(\text{male})} = \frac{e^{-44.45+0.66\cdot\text{height}}}{1 + e^{-44.45+0.66\cdot\text{height}}}$$

Making predictions

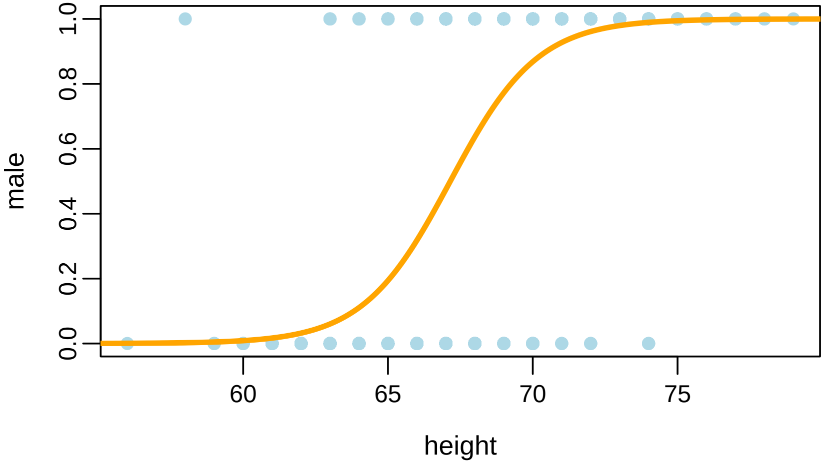
We can use `predict` to automate the process:

```
predict(model, data.frame(height=69), type="response")

   1
0.77
```

We predict that someone who is 5'9" has a 77% chance of being male.

Visualizing the model



Interpreting the coefficients

Our prediction equation is:

$$\log\left(\frac{P(\text{male})}{1-P(\text{male})}\right) = -44.45 + 0.66 \cdot \text{height}.$$

- When height increases by 1 inch, we predict that the log odds of being male will increase by 0.66.

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$$\log\left(\frac{P(\text{male})}{1-P(\text{male})}\right) = -44.45 + 0.66 \cdot \text{height}.$$

- When height increases by 1 inch, we predict that the log odds of being male will increase by 0.66.
- Equivalently: increasing height by 1 inch *multiplies* the odds by $e^{0.66} = 1.94$; i.e., increases the odds by 94%.

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- There are many "pseudo-R^2" metrics, but they are difficult to interpret.
- We'll focus on classification accuracy and the confusion matrix.

How many cases did we accurately predict?

Prediction rule: predict male if $\widehat{P(\text{male})} \geq 0.5$, else female.

```
predicted.sex <- ifelse(predict(model, type="response") >= 0
correct <- ifelse(predicted.sex == my.profiles$sex, 1, 0)
sum(correct) / nrow(my.profiles)

[1] 0.83
```

We correctly predict 83% of cases. Compare to predicting the majority class (male) for everyone: 60%.

The confusion matrix

Every case falls into one of four buckets:

		Actual	
		Positive	Negative
Predicted	Positive	TP	FP
	Negative	FN	TN

- **True positives** (TP): predicting male for someone who is male
- **False positives** (FP): predicting male for someone who is female
- **False negatives** (FN): predicting female for someone who is male

The confusion matrix

```
xtabs(~ predicted.sex + my.profiles$sex)

             my.profiles$sex
predicted.sex     f     m
            f 19466  5494
            m  4623 30243
```

5494 false negatives (true M predicted as F) and 4623 false positives (true F predicted as M).

Model evaluation: Final thoughts

- You can trade off false positives and negatives by using different cutoffs (predict 1 when $\hat{p} > c$ for $c \neq 0.5$).
- False positives and negatives can have very different costs (e.g., loan approval).
- These are **in-sample** measures—optimistic! Use train/test splits or cross-validation for out-of-sample accuracy.

Applications

- **Finance:** Predicting which customers are likely to default on a loan
- **Advertising:** Predicting response to a campaign
- **Marketing:** Predicting purchase or sign-up